

TrES-2b

R-0027

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_260320 · created 2026-07-08T14:33:38+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461119.9443 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.043
Transit depth (Rp/Rs)^2	1.82 +/- 1.16 [%]
Orbital Inclination (inc)	79.84 +/- 2.86
Airmass coefficient 1 (a1)	2.49 +/- 1.21
Airmass coefficient 2 (a2)	-0.54 +/- 0.16
Scatter in the residuals of the lightcurve fit is	49.85 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	7.51
Transit Duration (day)	0.016 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.135 ± 0.043 vs 0.1254 (deviation 0.22σ)
Duration fit vs published	0.016 d vs 0.0746 d (deviation 1.95σ)
Mid-time O–C	0.9 min
Depth SNR	3.1
Residual scatter	49.85 %

Light curve

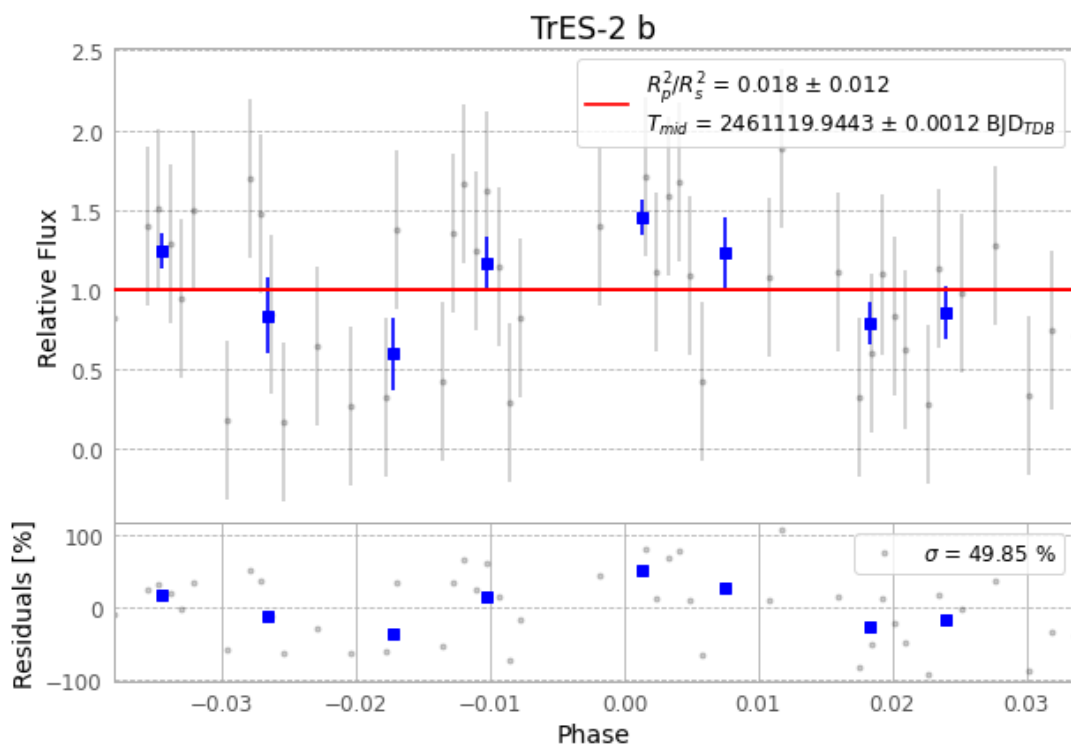


Figure 1 — FinalLightCurve_TrES-2 b_2026-03-20.png (SHA-256 386677bb57e3c57c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [276, 262], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 99a8b2269d2edaba...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

85 FITS frames (56 MB), TRES-2260320082415.FITS → TRES-2260320123916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-260320.log (d178ea02f1ed...), FinalLightCurve_TrES-2 b_2026-03-20.png (386677bb57e3...), FinalLightCurve_TrES-2 b_2026-03-20.csv (e20c1b59ce3c...)

Compute resources

Wall time 138.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.